

For this midterm, you are allowed to use your textbook (Hartle) and your personal notes. However, please refrain from using other sources, including the internet. An exception can be made for the starred (★) question. The questions marked with a (★) are required for students enrolled in 583 and bonus for students enrolled in 483.

1. Consider a 2-sphere with coordinates (θ, ϕ) and metric

$$ds^2 = d\theta^2 + \sin^2 \theta d\phi^2. \quad (1)$$

- (a) Show that lines of constant longitude ($\phi = \text{constant}$) are geodesics, and that the only line of constant latitude ($\theta = \text{constant}$) that is a geodesic is the equator ($\theta = \pi/2$).
- (b) Take a vector with components $V_\mu = (1, 0)$ and parallel-transport it once around a circle of constant latitude. What are the components of the resulting vector, as a function of θ ?
- (c) Consider the following metric for one space-like and one time-like dimension:

$$ds^2 = t^{-2}(-dt^2 + dx^2). \quad (2)$$

Find all the Christoffel symbols, and then calculate the geodesics of this spacetime. *Hint: for this second part, it is easiest to start with the definition of a geodesic as a maximum of proper time and find a relation for x and t by calculus of variations.*

2. Consider the following line element,

$$ds^2 = -\left(1 - \frac{2M}{\sqrt{r^2 + b^2}}\right) dt^2 + \left(1 - \frac{2M}{\sqrt{r^2 + b^2}}\right)^{-1} dr^2 + (r^2 + b^2)(d\theta^2 + \sin^2 \theta d\phi^2), \quad (3)$$

where M and b are positive, real constants. Note that we are using geometrical units where $G = c = 1$.

- (a) Write down the metric and the inverse metric in the (t, r, θ, ϕ) coordinate system.
- (b) Write down the non-zero Christoffel symbols $\Gamma_{\beta\gamma}^\alpha$ for this metric. *Hint: although there are 40 Christoffel symbols that are distinct by taking account of the symmetry under $\beta \leftrightarrow \gamma$, only 9 are non-zero.*
- (c) Write the geodesic equation in this metric for a massive particle, i.e. write equations for $\ddot{t}, \ddot{r}, \ddot{\theta}, \ddot{\phi}$ where the double-dots are with respect to proper time τ .
- (d) Calculate the effective potential for timelike orbits for arbitrary M and b .
- (e) Do bound orbital solutions exist for $M = 0$? Why or why not?
- (f) (★) Build a numerical solver for integrating the geodesic equation given an initial position ϵ and ℓ . Here, ϵ corresponds to energy density as written in class. Can you find any interesting geometries and/or trajectories?